



Training Program 1: Advanced NLP & Large Language Models (LLMs)

- M01: Text Preprocessing
- M02: Word and Sentence Embeddings
- M03: Transformer Architectures
- M04: Entity-Centric NLP
- M05: Applied NLP Tasks
- M06: Multimodal NLP
- M07: LLM Fundamentals
- M08: Prompt Engineering
- M09: Fine-Tuning & PEFT
- M10: RAG & Real-World Projects



Training Program 1: Advanced NLP & Large Language Models (LLMs)

1. **Why Vectors? (Foundations of Embeddings)**
2. **Word2Vec (CBOW & Skip-gram)**
3. **GloVe (Global Co-occurrence Models) fastText (Subword-aware Static Embeddings)**
4. **BERT and Contextual Embeddings**
5. **Sentence Embeddings (SBERT & Semantic Search)**
6. **Synthesis and Transition to Transformers**

M02 – Embeddings and Vector Representations

Chapter 2.1 – Introduction to Embeddings

- Embeddings map discrete symbols into numerical vectors
- Words and tokens become points in a continuous space
- Similar meanings are represented by nearby vectors
- This enables efficient mathematical processing of language



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Chapter 2.1 – Introduction to Embeddings

The Distributional Hypothesis

- Meaning emerges from usage and context
- Words appearing in similar contexts share meaning
- Models learn semantics without explicit definitions
- Context statistics drive semantic understanding



M02 – Embeddings and Vector Representations

Chapter 2.1 – Introduction to Embeddings

From One-Hot Encoding to Dense Vectors

- One-hot vectors are sparse and extremely high-dimensional
- All words are orthogonal and unrelated
- Dense embeddings reduce dimensionality
- Semantic similarity becomes measurable

Sparse One-Hot Vectors

0	0	0	0	1	0	1	0	0	0	0	1
0	1	0	0	0	0	1	0	0	0	0	0
0	0	1	0	0	0	0	1	0	0	0	0
0	0	0	1	0	0	0	0	1	0	0	0
0	1	0	0	0	1	0	0	0	0	0	0
0	0	0	0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	0	0	1	0	0
0	0	0	0	0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0	0	0	0	1



Dense Vectors

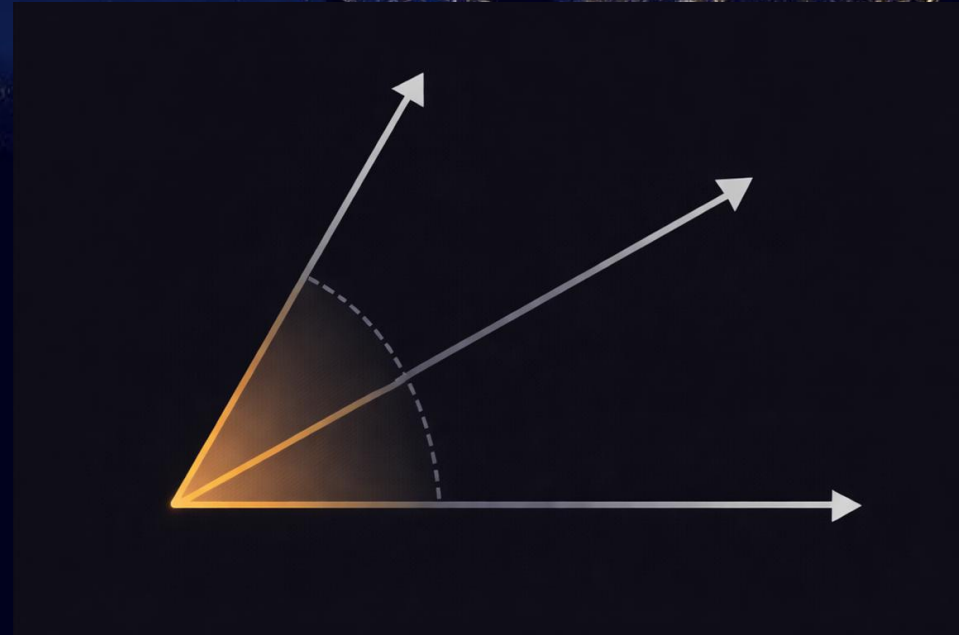
0.8	-0.2	
		-0.8
	0.6	
		-0.3

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Chapter 2.1 – Introduction to Embeddings

Measuring Semantic Similarity

- Vector similarity measures semantic closeness
- Cosine similarity compares vector orientation
- Similar meanings produce high similarity scores
- This enables search, clustering, and retrieval





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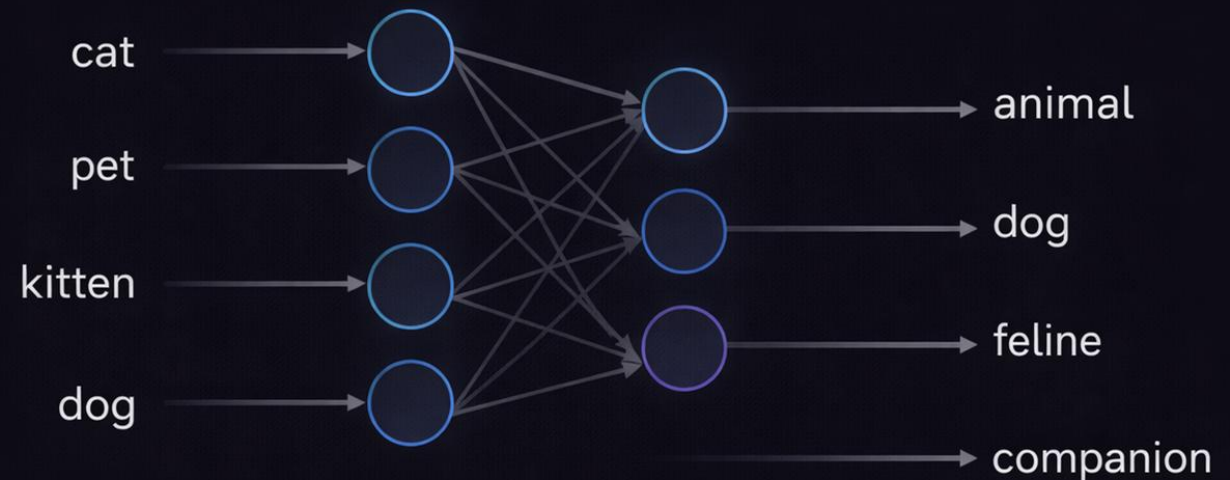
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Chapter 2.2 – Word2Vec

What Is Word2Vec?

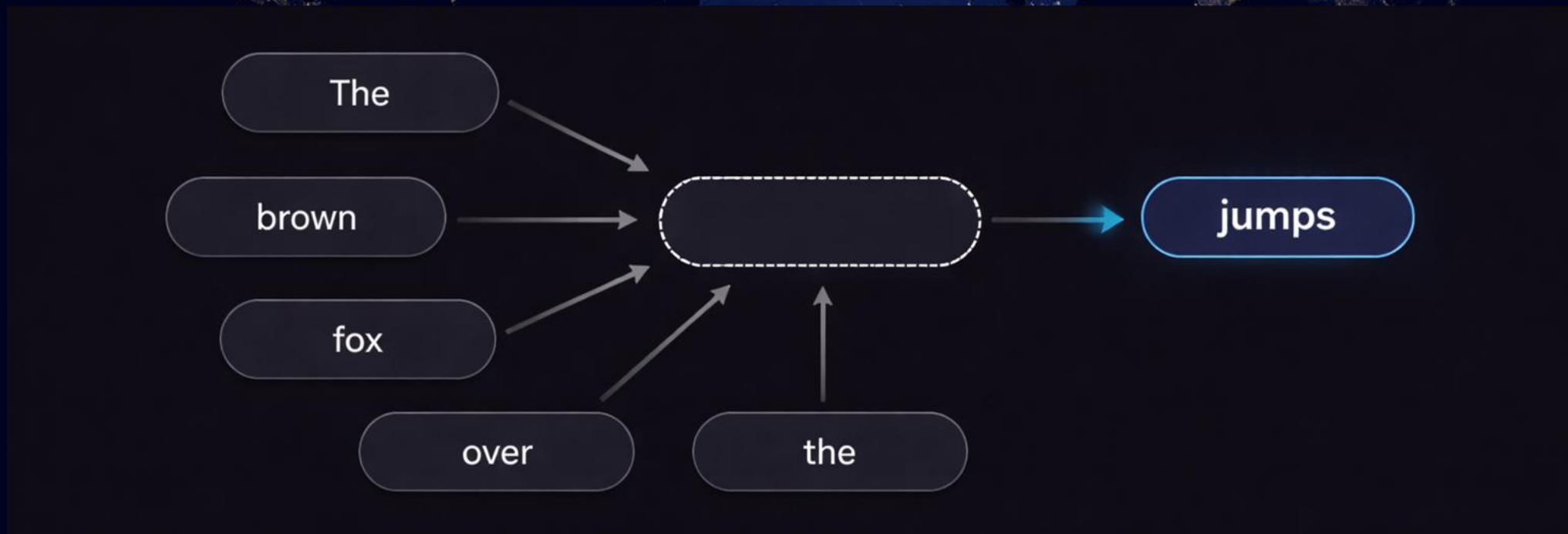
- Word2Vec learns word meanings from raw text
- It uses self-supervised learning from context
- Words are embedded into a shared vector space
- Meaning emerges from prediction tasks



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Chapter 2.2 – Word2Vec

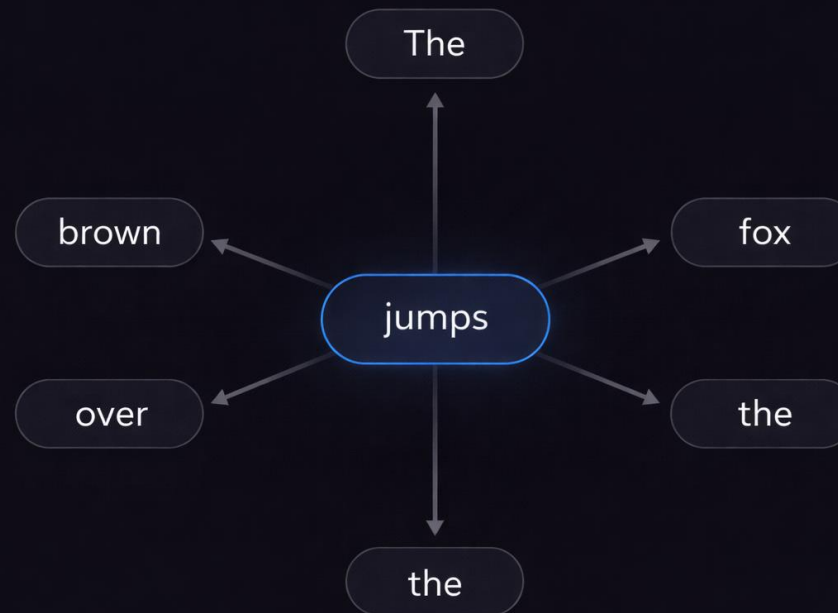
Continuous Bag-of-Words (CBOW)



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Chapter 2.2 – Word2Vec

Skip-gram Model



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Chapter 2.2 – Word2Vec

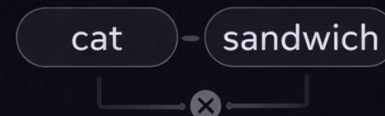
Negative Sampling and Word Analogies

- Full softmax is computationally expensive
- Negative sampling simplifies training
- Only a few negative examples are updated
- Word vectors capture linear semantic relations

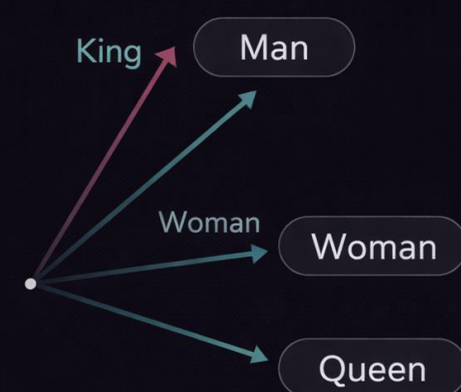
✓ Positive



✗ Negative



King – Man + Woman $\approx$ Queen





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Chapter 2.3 – GloVe: Global Vectors for Word Representation

Why Global Co-occurrence Matters

- Word2Vec learns from local context windows
- Some semantic patterns are global, not local
- Global co-occurrence captures corpus-wide statistics
- GloVe combines local and global information

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Chapter 2.3 – GloVe: Global Vectors for Word Representation

The GloVe Co-occurrence Matrix

- Word2Vec learns from local context windows
- Some semantic patterns are global, not local
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Chapter 2.3 – GloVe

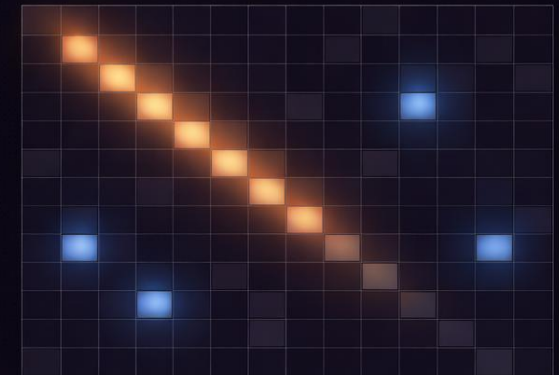
GloVe vs Word2Vec

- Word2Vec is prediction-based and local
- GloVe is count-based and global
- Both produce dense word embeddings
- GloVe often excels in similarity and analogy tasks

Streaming Local Context
(Word2Vec)

The brown fox → jumps

Global Matrix Optimization
(GloVe)





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Chapter 2.4 – fastText

Limitations of Static Word Embeddings

- Word2Vec and GloVe assign one vector per word
- Unseen words are mapped to an unknown token
- Morphological variations are poorly represented
- This limits robustness in real-world text

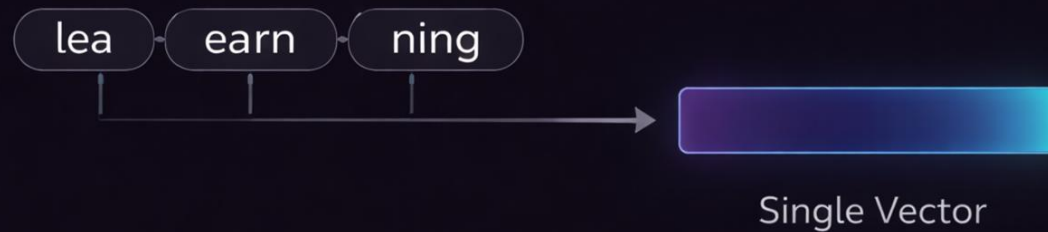
The brown fox jomps over the log.

<UNK>

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Chapter 2.4 – fastText

Character N-grams in fastText



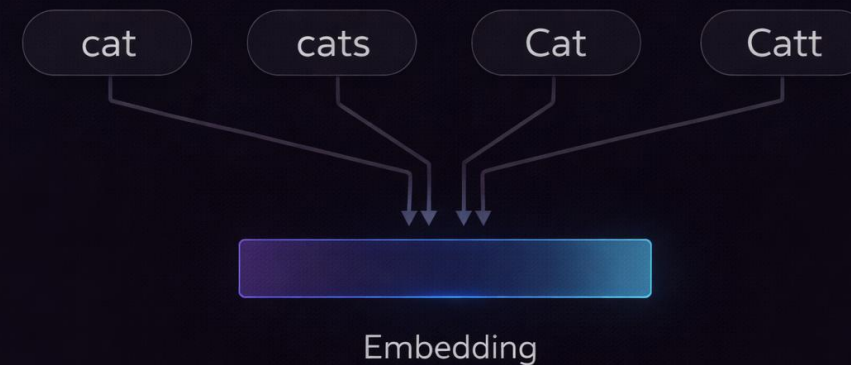
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Chapter 2.4 – fastText

Practical Advantages of fastText

- Strong performance on morphologically rich languages
- Robust to typos and spelling variations
- Fast training and low inference cost
- Widely used for large-scale text classification

Robustness & Efficiency





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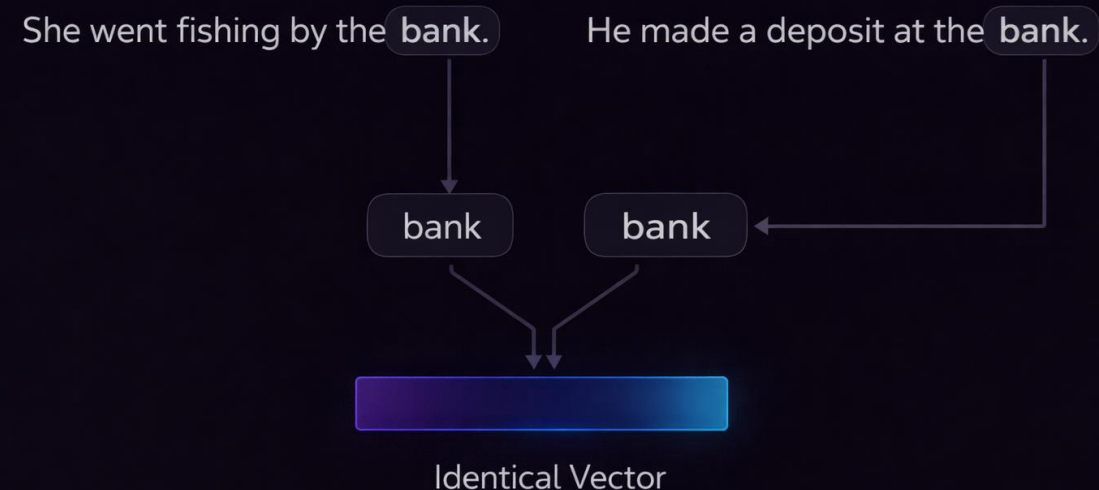
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Chapter 2.5 – BERT

Practical Advantages of fastText

- Static embeddings assign one vector per word
- Meaning does not change with context
- Polysemous words share a single representation
- This limits language understanding



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Chapter 2.5 – BERT

Contextual Embeddings with Transformers

She went fishing by the
bank, and he deposited
money in the bank.

Transformer
Block

bank

Word Vector

bank

Word Vector

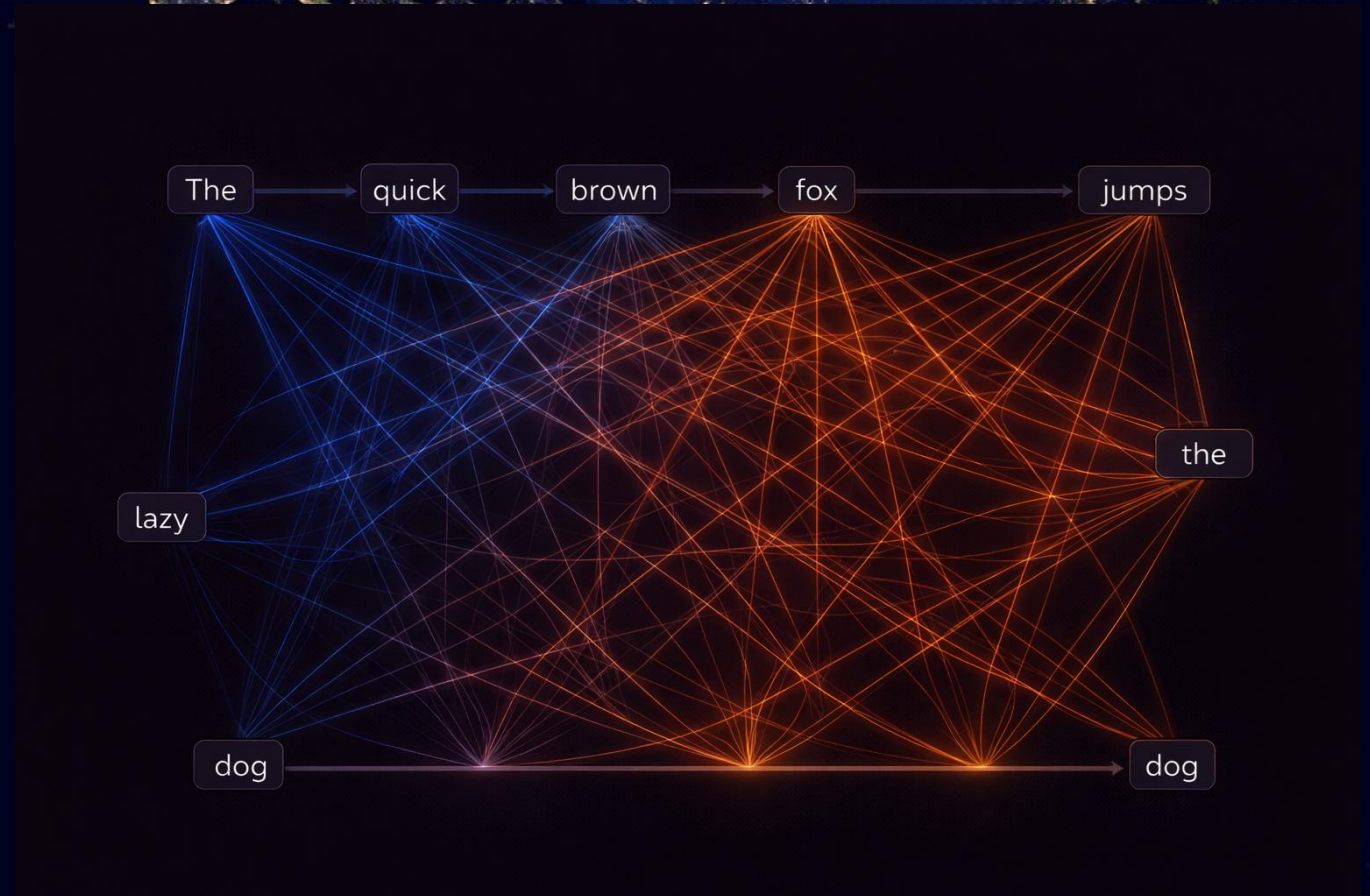
Same word produces different vectors

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Chapter 2.5 – BERT

Bidirectional Self-Attention

- BERT processes text in both directions
- Each token attends to all other tokens
- Context is built globally, not sequentially
- This enables deeper semantic understanding



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Chapter 2.5 – BERT

Pre-training and Fine-tuning

- Static embeddings assign one vector per word
- Meaning does not change with context
- Polysemous words share a single representation
- This limits language understanding

Large Pre-Training

General Purpose

Small Task-Specific

Fine Tuning



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Chapter 2.6 – Sentence Embeddings

Why Token-Level Embeddings Are Not Enough

The quick brown fox jumps over the lazy dog.



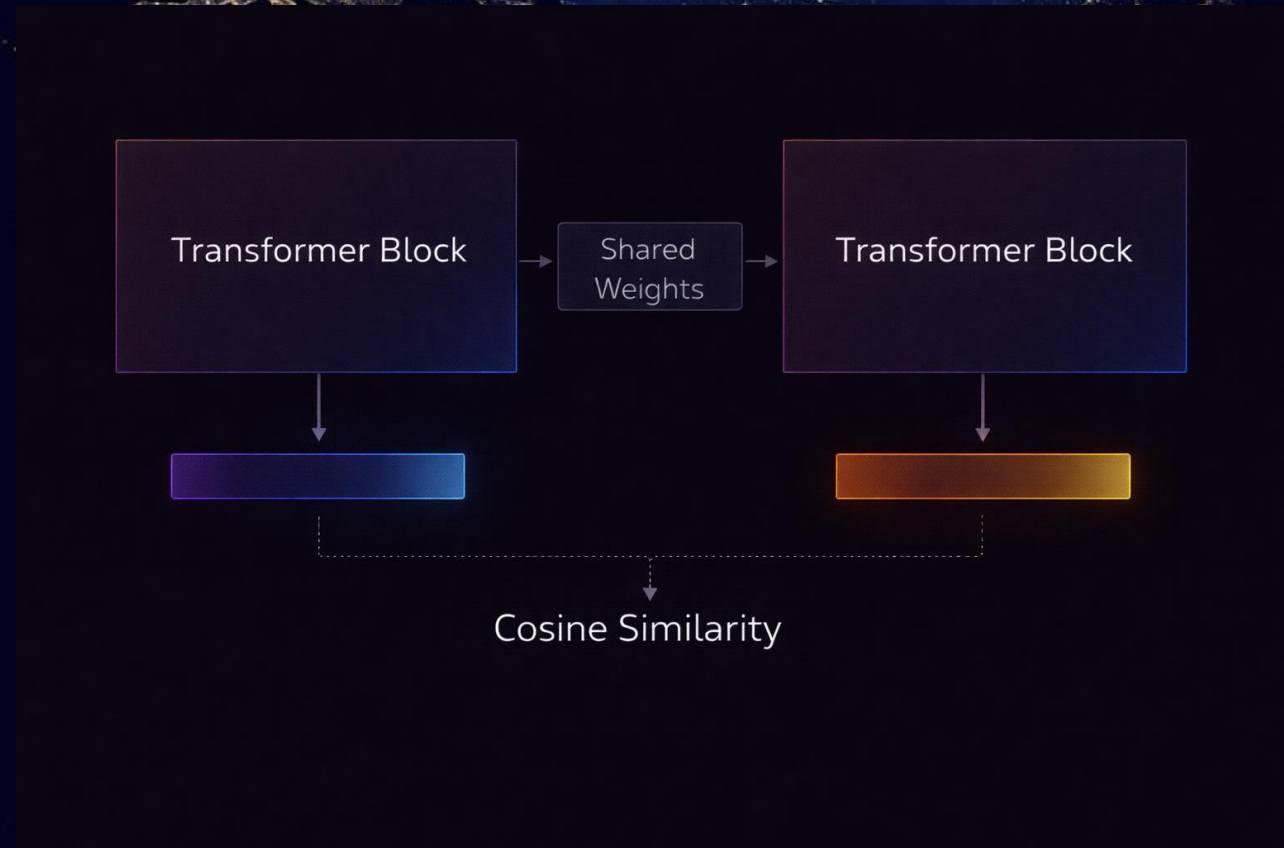
No clear single representation

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Chapter 2.6 – Sentence Embeddings

Sentence-BERT (SBERT) Architecture

- SBERT uses a Siamese transformer architecture
- Sentences are encoded independently
- Vector comparison uses cosine similarity
- Similarity computation is fast and scalable

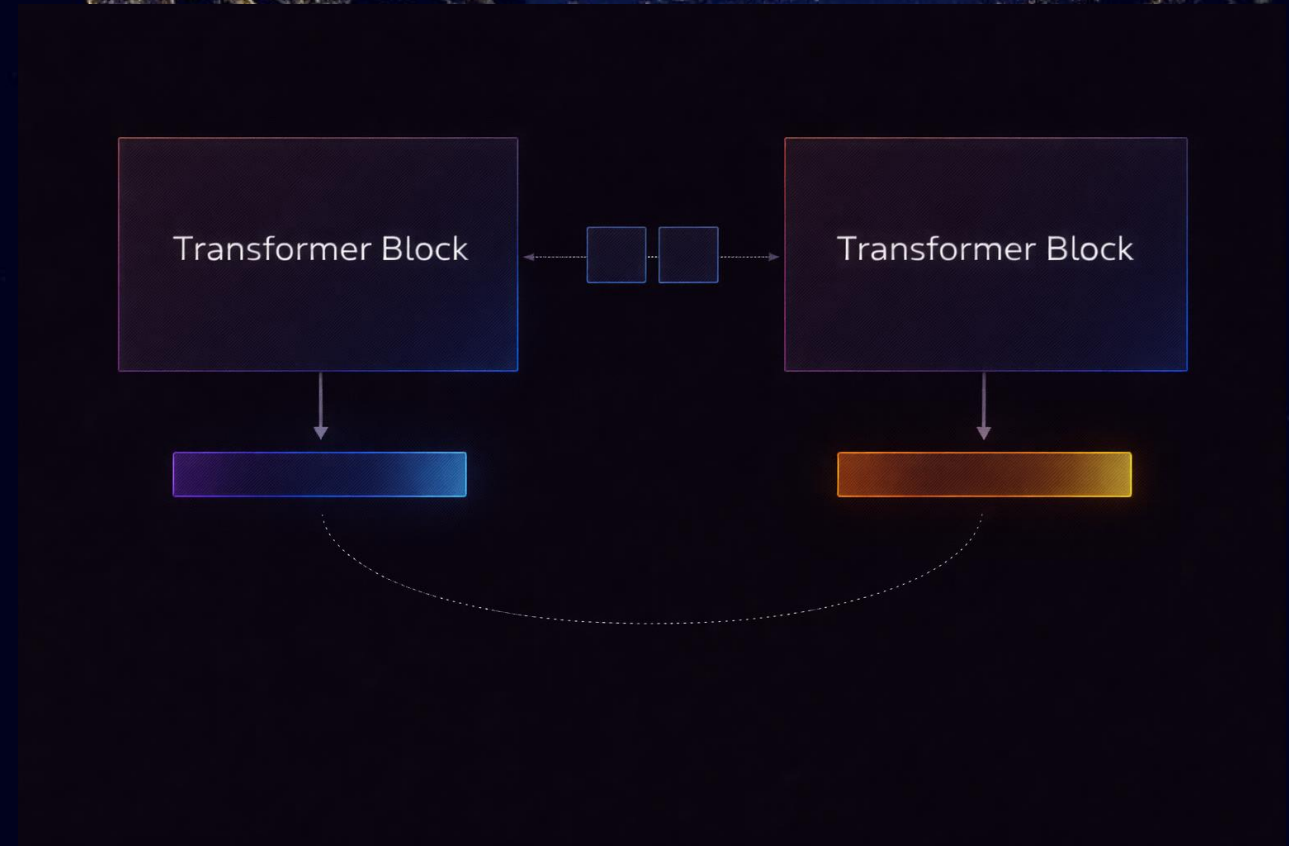


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Chapter 2.6 – Sentence Embeddings

Sentence-BERT (SBERT) Architecture

- Semantic search over large document collections
- Retrieval-Augmented Generation (RAG)
- Clustering and topic discovery
- Near-duplicate detection





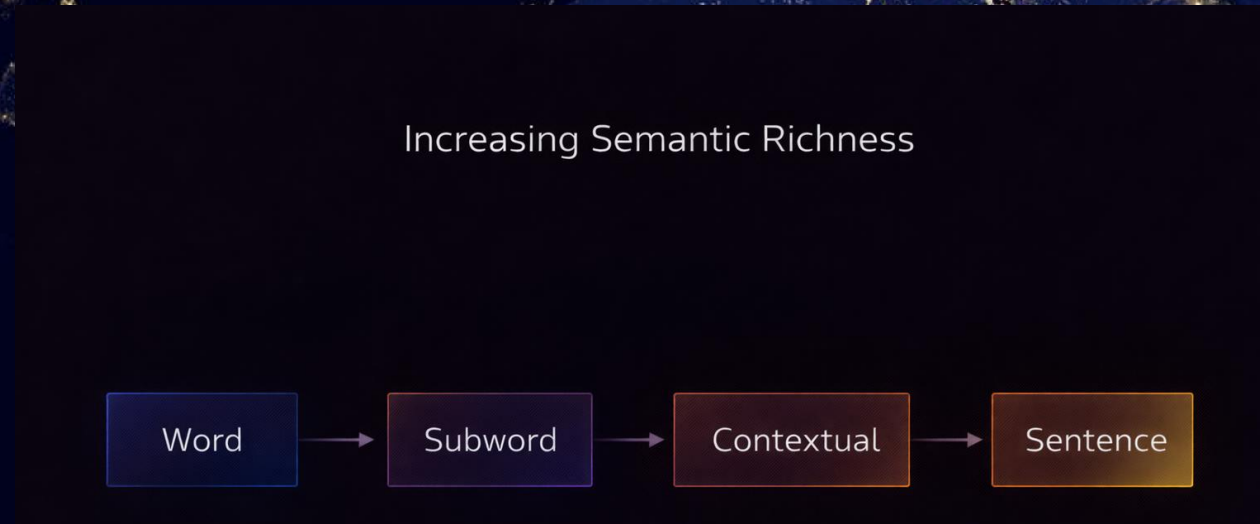
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Chapter 2.7 – Synthesis of Embedding Methods

From Static to Contextual Embeddings



- Static embeddings map words to fixed vectors
- Subword methods improve robustness and coverage
- Contextual models adapt meaning to usage
- Sentence embeddings enable scalable semantics

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Chapter 2.7 – Synthesis of Embedding Methods

Why Transformers Are the Next Step

- Embeddings alone do not perform reasoning
- Transformers process vectors through attention
- Meaning emerges from interaction, not isolation
- Architecture defines intelligence

